

Differential Uniformity, One Closure Law: the Frobenius Twist and the Kasami Even- k Move

Abstract

Differential uniformity $\delta(f)$ measures how well an S-box resists differential cryptanalysis. The two best classes are APN ($\delta = 2$) and PN ($\delta = 1$).

This pearl rests on one law: *relabelling the output of f by an additive bijection leaves δ unchanged*. From it we get two payoffs, almost for free:

- the **Frobenius twist** — raising the output to a power of two keeps APN;
- the **Kasami even- k move** — the classical odd- k APN theorem implies the even- k case, with no new analysis.

Every boxed statement is checked in Lean 4 / Mathlib. One law about addition does all the work.

1 Introduction

Differential cryptanalysis attacks a cipher by feeding it input pairs that differ by a fixed amount a , then watching which output difference b shows up most often. An S-box that smears differences out evenly is hard to attack.

The measure is the *differential uniformity* $\delta(f)$: the most input pairs that share any one nonzero input difference and output difference. Smaller is better. Two classes sit at the bottom:

$$\text{APN: } \delta(f) = 2 \quad (\text{char } 2), \quad \text{PN: } \delta(f) = 1 \quad (\text{odd char}).$$

Power functions $x \mapsto x^d$ supply most known APN families: Gold, Kasami, Welch, Niho, inverse, Dobbertin.

Big idea. We do not re-prove a hard APN theorem. We isolate *one* elementary law — δ ignores additive relabelling of the output — and show how much it buys. The hard classical input (odd- k Kasami) is a black box; the new content is the *reduction*.

The trick is to state the definition with *no field structure*, using only $+$ and $-$. Then the closure law is one line, and the Frobenius corollary is just a special case.

2 The invariant

Push f sideways by a step a and see how much it moved:

$$\text{derivative of } f \text{ in direction } a : \quad f(x+a) - f(x).$$

Fix a nonzero step a and a target b , then count the hits:

$$\text{fiberCard } f \ a \ b = \#\{x : f(x+a) - f(x) = b\}.$$

Take the worst case over all $a \neq 0$ and all b :

$$\delta(f) = \max_{a \neq 0, b} \text{fiberCard } f \ a \ b.$$

Flat differences mean small δ , which means hard to attack:

$$\boxed{\text{IsAPN } f : \iff \delta(f) = 2}$$

$$\boxed{\text{IsPN } f : \iff \delta(f) = 1}$$

The definition uses only $+$ and $-$, never multiplication. So we state it for *any* finite additive groups V, W and $f : V \rightarrow W$ — no field, no characteristic. That generality is what makes the closure law free.

2.1 The primitive, in Lean

```
def derivMap (f : V → W) (a x : V) : W := f (x + a) - f x

def fiberCard (f : V → W) (a : V) (b : W) : ℕ :=
  (univ.filter fun x => derivMap f a x = b).card

def differentialUniformity (f : V → W) : ℕ :=
  (univ.filter fun a => a ≠ 0).sup
    fun a => univ.sup fun b => fiberCard f a b

def IsAPN (f : V → W) : Prop := differentialUniformity f = 2
```

Four lines, and V, W are any finite additive groups. Using `Finset.sup` keeps δ a natural number, so the closure-law rewrites stay purely combinatorial.

Cool fact. In characteristic two, “ $\delta(f) \leq 2$ ” has a hands-on shape: among any three inputs sharing a derivative value, two must coincide. “At most two-to-one.”

Theorem 1 (collision form). *If $\text{derivMap } f \ a \ x = \text{derivMap } f \ a \ y = \text{derivMap } f \ a \ z$, then $x = y$,*

$x = z$, or $y = z$.

The two descriptions are proved equal in Lean (`isAtMostTwoToOne_iff_diffUnif_le_two`).

2.2 A worked example

Take $F = \text{GF}(8)$ and the Gold map $f(x) = x^3$ (the smallest Kasami exponent, $d_1 = 2^2 - 2^1 + 1$). Fix $a \neq 0$. In characteristic two,

$$f(x+a) - f(x) = (x+a)^3 - x^3 = a(x^2 + ax) + a^3.$$

For a target b , the equation $f(x+a) - f(x) = b$ is a quadratic in x . Over characteristic two such a quadratic has at most two roots, and $x \mapsto x+a$ leaves it fixed — so roots come in pairs $\{x, x+a\}$.

Every nonempty fibre therefore has exactly two elements: $\text{fiberCard } f \ a \ b \in \{0, 2\}$, so $\delta(f) = 2$. This is the base case the Kasami reduction rests on.

3 The one closure law

This is the heart of the paper. Let σ be an *additive bijection* of the output (an element of $\mathcal{W} \simeq+ \mathcal{W}'$): a relabelling that respects $+$.

Theorem 2 (closure law). $\delta(\sigma \circ f) = \delta(f)$.

Relabelling the output cannot change how spread the differences are. Here is the whole reason. First, σ slides through the derivative:

$$\text{derivMap } (\sigma \circ f) \ a \ x = \sigma(\text{derivMap } f \ a \ x),$$

because $\sigma(u-v) = \sigma(u) - \sigma(v)$. Second, it just renames each fibre one-for-one:

$$\text{fiberCard } (\sigma \circ f) \ a \ b = \text{fiberCard } f \ a \ (\sigma^{-1}b).$$

Same counts, shuffled. Same maximum.

```

lemma derivMap_comp_addEquiv (f : V → W) σ( : W ≃+ W') (a x : V) :
  derivMap σ( of) a x = σ(derivMap f a x)

lemma fiberCard_comp_addEquiv (f : V → W) σ( : W ≃+ W') (a) (b) :
  fiberCard σ( of) a b = fiberCard f a σ(.symm b)

lemma differentialUniformity_comp_addEquiv
  (f : V → W) σ( : W ≃+ W') :
  differentialUniformity σ( of) = differentialUniformity f

```

```

theorem IsAPN.comp_addEquiv
  {f : V → W} (hf : IsAPN f) (σ : W ≃+ W') : IsAPN σ( ∘ f)

```

Big idea. One move carries a whole equivalence class. Prove a single example APN, and get every additive relabelling for free. The law never sees that W might be a field.

4 Payoff one: the Frobenius twist

Now work in a finite field F of characteristic two. The Frobenius power $\sigma : x \mapsto x^{2^j}$ is

- **additive:** $(x + y)^{2^j} = x^{2^j} + y^{2^j}$ (the freshman’s dream, iterated);
- **bijective:** an injective endomorphism of a finite field.

So it is just one instance of the closure law.

Corollary 1. *If f is APN, then $x \mapsto f(x)^{2^j}$ is APN.*

```

theorem IsAPN.comp_frobenius
  {f : F → F} (hf : IsAPN f) (j : ℕ) :
  IsAPN (fun x => (f x) ^ (2 ^ j))

```

The proof: build σ from Mathlib’s `iterateFrobenius`, then apply `IsAPN.comp_addEquiv`. Raising the output to a power of two is invisible to APN-ness — and that freedom drives the second payoff.

5 Payoff two: the Kasami even- k move

The Kasami power function on $\text{GF}(2^n)$ is $x \mapsto x^{d_k}$,

$$d_k = 2^{2k} - 2^k + 1.$$

Classically it is APN whenever $\gcd(k, n) = 1$, proved via the *odd* k case. We handle even k not by new analysis, but by *moving* the odd case onto it.

First, restate the law as a plain bijection, and specialise σ to a Frobenius power on $f = x^d$:

```

lemma apn_comp_additive_bijective
  {f : F → F} (hf : IsAPN f) σ
  { : F → F} (σh_bij : Function.Bijective σ)

```

```
(σh_add : ∀x y, σ(x + y) = σx + σy) : IsAPN σ( of)
```

```
lemma apn_frob_twist
```

```
{d : ℕ} (hd : IsAPN (fun x : F => x ^ d)) (j : ℕ) :  
IsAPN (fun x : F => x ^ (d * 2 ^ j))
```

Multiplying the *exponent* by 2^j keeps APN, since $x^{d \cdot 2^j} = (x^d)^{2^j}$ is a Frobenius twist.

5.1 The key congruence

On $\text{GF}(2^n)$ the unit group has order $2^n - 1$, so for $x \neq 0$ exponents only matter modulo $2^n - 1$ (and $x = 0$ is fixed by every power map). The Kasami exponents at k and at its complement $n - k$ are linked:

Lemma 1 (exponent congruence). *For $0 < k < n$, $d_k \equiv d_{n-k} \cdot 2^{2k} \pmod{2^n - 1}$.*

A short computation:

$$d_{n-k} \cdot 2^{2k} = (2^{2(n-k)} - 2^{n-k} + 1) 2^{2k} = 2^{2n} - 2^{n+k} + 2^{2k}.$$

Reduce $2^{2n} \equiv 2^n \equiv 1$ and $2^{n+k} \equiv 2^k$ modulo $2^n - 1$, and what remains is $2^{2k} - 2^k + 1 = d_k$. So on the field,

$$x^{d_k} = (x^{d_{n-k}})^{2^{2k}} :$$

one map, written either as a Kasami exponent or as a Frobenius twist of another (kasami_pow_frob_identity).

Lemma 2 (APN of the complement). *If $x^{d_{n-k}}$ is APN, then x^{d_k} is APN.*

```
lemma kasami_apn_of_complement
```

```
(hn : Fintype.card F = 2 ^ n)  
(k : ℕ) (hk : 0 < k) (hkn : k < n)  
(hpn : IsAPN (fun x : F => x ^ kasamiExp (n - k))) :  
IsAPN (fun x : F => x ^ kasamiExp k)
```

5.2 Parity flips for free

Why is the complement the right partner? Let n be *odd* and k be *even* with $\gcd(k, n) = 1$. Then:

- $n - k = \text{odd} - \text{even}$ is *odd*;
- coprimality survives: $\gcd(n - k, n) = \gcd(k, n) = 1$.

So $x^{d_{n-k}}$ is covered by the classical odd theorem, and the complement trick lifts it back to x^{d_k} .

Cool fact. The even case is the odd case seen through Frobenius. The parity barrier simply dissolves — no second analytic proof needed.

```

theorem kasami_is_apn_even_k
  (hn : Fintype.card F = 2 ^ n) (k : ℕ)
  (hk : 1 < k) (hk_even : Even k) (hkn : k < n)
  (hn_odd : Odd n) (hcop : Nat.Coprime k n)
  (hnk : n - k ≥ 2) :
  IsAPN (fun x : F => x ^ kasamiExp k)

theorem kasami_is_apn_general
  (hn : Fintype.card F = 2 ^ n) (k : ℕ)
  (hk : 1 < k) (hkn : k < n)
  (hn_odd : Odd n) (hcop : Nat.Coprime k n) :
  IsAPN (fun x : F => x ^ kasamiExp k)

```

The edge $k = n - 1$ lands on the Gold exponent $d_1 = 3$, proved APN directly. With it, `kasami_is_apn_general` covers every k coprime to n , with no parity restriction left.

6 The bridge back

We started field-free; we end where the textbooks are.

Theorem 3. *In characteristic two, if $\delta(f) \leq 2$ then for $a \neq 0$,*

$$f(x + a) - f(x) = f(y + a) - f(y) \implies y = x \text{ or } y = x + a.$$

Nothing is lost: the classical collision statement is *derived*, not assumed (`atMostTwoToOne_charTwo_collision`). The group definition and the field-theoretic folklore coincide exactly.

7 Formalization notes

A few choices keep things short and reusable:

- **δ over groups, not fields.** The primitive uses only $+$ and $-$, so the closure law is a fact about `AddEquiv` and Frobenius is a special case. The field enters only to make Frobenius additive and bijective.

- **δ valued in $\mathbb{N}$.** Using `Finset.sup` of `fiberCard` makes APN-ness a decidable equation and fibre re-indexing a `Finset` bijection.
- **Exponents mod $2^n - 1$.** The Kasami identity is proved as an equality of functions on F (split on $x = 0$, use the order of $F^\times$), avoiding congruence side-conditions.
- **Odd case as a hypothesis.** The even- k theorem takes the odd- k result as input, so it is robust to whichever proof of the odd case is supplied.

The files are `DifferentialUniformity.lean` (definition, collision form, closure law, Frobenius twist) and `KasamiEvenK.lean` (congruence, complement trick, even- k and general theorems).

8 Related work

APN functions come from differential cryptanalysis and have a large literature of power-function families: Kasami $2^{2k} - 2^k + 1$ with $\gcd(k, n) = 1$, Gold $2^k + 1$, and others. The odd- k APN property is textbook; the even- k reduction through the complement is folklore. Our contribution is to make that reduction fully explicit and machine-checked, with the closure law as its only structural ingredient. Mathlib supplies the finite-field and Frobenius infrastructure (`frobenius`, `iterateFrobenius`, order of the unit group); what is new is the small, self-contained theory of δ , stated for arbitrary finite additive groups.

9 Conclusion

The take-away is methodological. One almost trivial law — δ is blind to additive relabelling of the output — is enough to

1. record the Frobenius twist for free, and
2. collapse the parity split in Kasami to one exponent congruence plus the same law.

Stating the invariant at the right level of generality, with no field structure, is what makes both payoffs mechanical.

Big idea. Big idea, small footprint: one law about addition does all the work.